Advanced Technologies: Developing Mood Responsive Web Application

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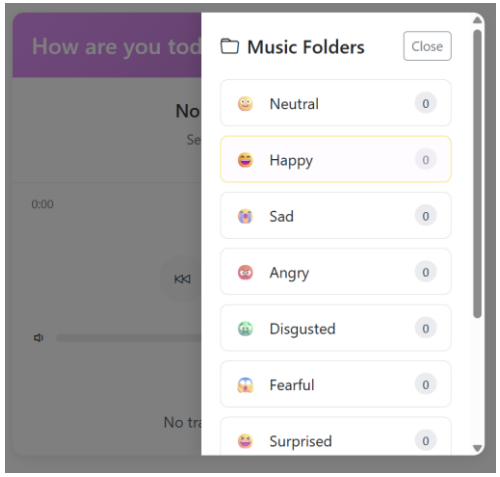


Figure1. screenshot of part of interface

Abstract
There are different ways to detect human's mood, including through webcam, voice, wearable sensors such as smartwatch and even more ways. These can identify human's feelings, like joy, anger and sadness. This project aims to hack empathy into user experience (UX), by developing a user interface that modifies layout, colour palette and app's control in response to user's emotional state.

Author Keywords
User experience (UX), User interface (UI), music, emotion/mood, progressive web application (PWA)

Introduction
Music recommendation systems are used with the aim of actively providing users with content of high interests based on behaviour, purposes, or past experiences (de Assunção and Zaina, 2022). This project was developed in response to project's request to explore how user's emotion state can be utilised to tune application's content and visual style to create a more responsive and empathetic user experience.

Building upon the foundations of affective computing and adaptive interface design, this project integrates real-time facial emotion recognition with a functional music player. Using the ml5.js library and the face-

api.js model, the system detects seven core emotions which are neutral, happy, sad, angry, disgusted, fearful, and surprised, and maps them to thematic colour schemes and corresponding music folders. The interface smoothly changes by feelings, while playing logic bypasses the mood based changing folder without interruption. This logic personalises music selection and modifies the visual environment dynamically and responds in real-time, thus embodying emotion-responsive design as proposed by recent frameworks of adaptive affective interfaces (Alipour, Éric Céret & Dupuy-Chessa, 2023). The developed application illustrates the power of emotion-aware systems in increasing user engagement via transparent, context-sensitive adjustments.

Background Research

As part of the development of an emotion-responsive music player, research was done in three distinct areas: colour and emotion, the effect on UI to UX, and the relationship between emotion and music.

The visual side of the interface and the interaction through the user such as layout, buttons, and colour schemes are the main areas of focus of UI design. The primary aim is to come up with an interface that is not only appealing, but also easy to comprehend and straightforward to use (Hamidli, 2023). Nevertheless, good design is slowly transitioning beyond just being user-friendly. A paper by Willis (2007) have claimed that the following stage is to develop "emotionally intelligent" interfaces. These systems would attempt to figure out the user's emotions and then change the way they interact with the user to elevate the user experience. This means that, besides usability, another design goal is to establish a positive emotional connection with the user.

Colour plays a very significant role in UI design. The colours chosen for an interface do not only beautify it; they can also affect the user's mood and even change their behaviour. colours serve to clarify information, they focus on the most important parts of the text, and they lead the user through the system (Hamidli, 2023). This concept is called colour psychology. To give an example, Ouyang (2021) concluded that implementing colour based on psychology is a vital part of good UX design, especially for critical systems such as healthcare websites. In such cases, correct colours not only make the system more user-friendly and provide clearer information but also, they can make the users feel more at ease. This proves that deciding on colours is a vital design decision rather than just a matter of style.

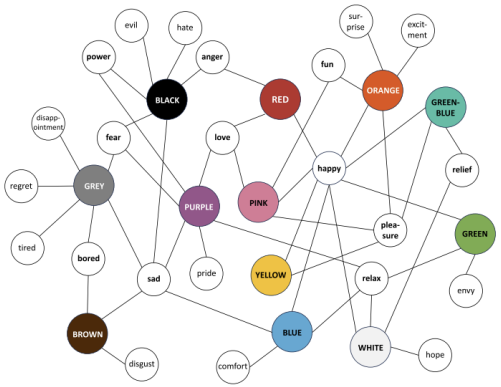


Figure2. A visual representation of the most frequent correspondences between affective concepts and 12 colour categories (Jonauskaite and Mohr, 2025)

The connection between music and emotion is complicated and has been extensively debated. There is no doubt that music is highly influential in our lives. In fact, playing suitable music helps people to relax, feel less stressed, have good sleep, and become more lovable and productive (Bakariya et al., 2023). Considering the strong impact that music has on the human brain, the demand for personal music services is on the rise. One recent study shows that there is a strong link between the current emotional state of a person and the kind of music that person would like to listen to (Faye et al., 2025). This points to the necessity of an emotionally intelligent personal music service that considers the current emotional state of the listener before making any recommendations.

Each of the studies on UI/UX, colour psychology, and music would, on their own, provide a stable basis. In addition, the real power for an engaging user experience lies in their coming together. The future of user interfaces lies in their being not just functional but also aware and adaptable to the user's needs - a theme central to emotive design (Willis, 2007). Such adjustment demands a definite visual language that colour psychology delivers by establishing scientifically supported links between colours and emotional states, thereby demonstrating that colour is a practical means of communication (Ouyang, 2021). As a matter of fact, the extensively researched ability of music to influence one's mood (Brijesh Bakariya et al., 2023) and the users' call for personalization (Faye et al., 2025) make the need for the audio part to be equally adaptable, quite evident.

Presently, systems only cater to one aspect of this loop. For instance, a music app may offer playlists based on a fixed "mood" label, or a website might utilize a predetermined colour scheme to bring about a certain feeling. There are only a few systems that attempt to close the feedback loop by using a live measure of the user's emotional state to dynamically synchronize both the visual theme and the auditory content in real time. This project is intended to fill that void. It installs a system where the analysis of the facial expression identifies the user's emotion. This emotion then becomes the driving force that connects the system's two main features. The colour of the user interface theme changes in real-time with respect to a previously established emotional palette. In addition, the system picks out the music folder that matches the detected emotion and takes the user towards a supplementary listening experience that is customized for him/her. It is quite important to note that the songs in these folders are uploaded by the user, so that there can be a high level of personal selection within the emotion-driven framework. Hence, this project is not just about separately implementing the three areas of research; it combines them into one single, interactive, proof-of-concept for more holistic and empathetic human-computer interaction.

Methodology

Being faced with skill constraints, especially in the absence of experience with using Python. I took a web-first approach using more inclusive technologies. The application was developed using basic web development technologies included HTML, CSS, and JavaScript, and then converted into a Progressive Web Application using instructions found on a YouTube tutorial to support offline access and mobile installation.

Bootstrap 5 was used for designing user interface to create a uniform UX experience on both desktop and mobile platforms. For the main part, which is application of emotion detection, I utilized existing resources for applications of creative programming using the p5.js library for video recording features, incorporating the pre-trained API for facial expressions using the ml5.js library to support complex emotion detection without the need for training a machine learning model.

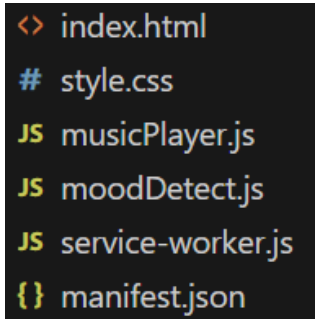


Figure3. Screenshot of VS code, methods used to build prototype

Implementation

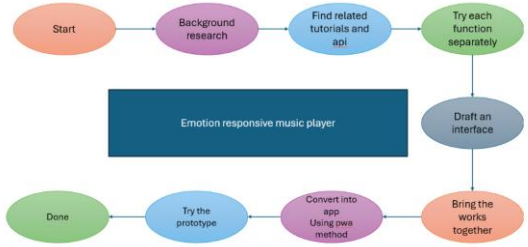
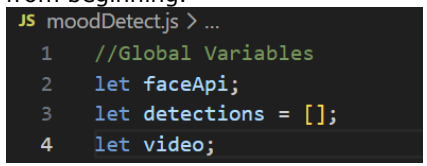


Figure4. Progress of the project

The final prototype built with visual studio code, and it is a web application.
Following techniques used in project:

1. Mood detection

Mood detection used the ml5.js library's pre-trained Face-API model found from p5.js, which provides real-time facial expression analysis without build the system from beginning.



```
JS moodDetect.js > ...
//Global Variables
let faceApi;
let detections = [];
let video;
```

Figure5. Using api

The use of mood detection:

- changing theme colour

```
//UI transition
function colourTransition() {
  if (currentEmotion !== targetEmotion) {
    transitionProgress += TRANSITION_SPEED;
    if (transitionProgress >= 1) {
      transitionProgress = 1;
      currentEmotion = targetEmotion;
    }
    const currentColour = emotionColours[currentEmotion];
    const targetColour = emotionColours[targetEmotion];

    if (transitionProgress < 1) {
      const blendedColour = interpolateColour(currentColour, targetColour, transitionProgress);
      updateUITheme(blendedColour, targetEmotion);
    } else {
      updateUITheme(targetColour, targetEmotion);
    }
  }
}

function updateUITheme(colour, emotion) {
  document.documentElement.style.setProperty('--theme-colour', colour);
  document.documentElement.style.setProperty('--theme-emotion', emotion);
}
```

Figure6. Using mood to change theme colour

Once the face api model identifies a dominant facial expression, the system associates it with a corresponding colour palette according to the psychological associations research (Jonauskaite & Mohr, 2025). For example, joy results in a vivid yellow decor, whereas grief causes a dull grey. This colour alteration is done via CSS custom properties "theme-colour", which are shared throughout the interface, thus the player header, progress bars, buttons, and highlighting elements get the colour from the changed

interface. The transitions are animated via CSS easing functions over 0.5 seconds to ensure that the changes are smooth and non-disruptive.

- switching music folder

```
//Select folder based on emotion
function triggerFolderChangeForEmotion(emotion) {
  const folderId = emotionToFolderMap[emotion];

  if (folderId && typeof window.selectFolder === 'function') {
    const currentFolder = window.currentFolderId || 1;

    if (folderId !== currentFolder) {
      const isMusicPlaying = window.isPlaying || false;

      if (!isMusicPlaying) {
        window.selectFolder(folderId);
      }
    }
  }
}
```

Figure7. Folder switched based on detected mood

The mood that has been detected decides the music folder through a folder management system. Each emotion is associated with a music folder named after the emotion and filled with user-uploaded mp3 files. When an emotion is newly and consistently detected, the application changes to the corresponding folder thereby updating the playlist interface instantly. Besides, this folder change is being silently held back during the ongoing audio playback and is activated only when playback stops.

2. Uploading file through web

```
//Upload file
dragDropArea.addEventListener('dragover', (e) => {
    e.preventDefault();
    dragDropArea.classList.add('dragover');
});

dragDropArea.addEventListener('dragleave', () => {
    dragDropArea.classList.remove('dragover');
});

dragDropArea.addEventListener('drop', handleFileDrop);

dragDropArea.addEventListener('click', () => {
    if (currentFolderId === null) selectFolder(1);
    fileInput.click();
});
```

Figure8. File upload section

The file upload interface, designed by following the tutorial from the YouTube video, "How To Create Drag and Drop Image Uploader using HTML, CSS, and JavaScript" (GreatStack, 2023). To upload the files, users can rely on the drag-and-drop method, where files can be uploaded through upload area.

3. Tracking the mp3 data

```
fileInput.addEventListener('change', async (e) => {
    const files = e.target.files;
    for (let file of files) {
        if (file.type.startswith('audio/')) {
            const fileName = file.name.replace(/\.[^/.]+$/, "");
            const duration = await getAudioDuration(file);
            const durationFormatted = formatTime(duration);
            addTrackToFolder(currentFolderId, file, fileName, "Unknown Artist", durationFormatted);
        }
    }
    fileInput.value = '';
    updatePlaylist();
});
}
```

Figure9. using asynchronous decodeAudioData() method

It integrates the process for track duration extraction via the asynchronous decodeAudioData() method available within the Web Audio API. Uploading an MP3 file trigger processing its data to obtain the exact audio duration in the number of seconds, which is then displayed in the m:s format through the custom function formatTime(), displaying a value such as 65s is 01:05.

4. Volume control slider

The volume controls are achieved by utilizing the range input with a custom HTML5 styled element coupled with the use of the Bootstrap5 form-range class to promote consistent styling across the browser. The theme colours for the slider's display change according to the detected emotional state with an automatic updating of the accent-colour CSS property linked to the current theme colours declared using the CSS custom properties. The JavaScript triggers changes in the volume level for the HTML5 Audio element immediately.

5. Building progressive web application

```
manifest.json > ...
{
  "name": "Emotion Responsive Music Player",
  "short_name": "MusicPlayer",
  "description": "Music player that responds to your emotions",
  "start_url": "/",
  "display": "standalone",
  "background_colour": "#4a6fa5",
  "theme_colour": "#4a6fa5",
  "icons": [
    {
      "src": "musical-note.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "musical-note.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

```
service-worker.js > ...
const CACHE_NAME = 'musicplayer-v1';
const urlsToCache = [
  '/',
  '/index.html',
  '/style.css',
  '/musicPlayer.js',
  '/moodDetect.js',
  '/manifest.json',
  '/musical-note.png'
];

self.addEventListener('install', event => {
  event.waitUntil(
    caches.open(CACHE_NAME)
      .then(cache => cache.addAll(urlsToCache))
  );
});

self.addEventListener('fetch', event => {
  event.respondWith(
    caches.match(event.request)
      .then(response => response || fetch(event.request))
  );
});
```

Figure10&11. Building progressive web application (PWA)

Followed tutorial from YouTube, “Progressive Web Apps in 100 Seconds // Build a PWA from Scratch” by Fireship (2020).

Findings

Apart from watching and reading tutorials to build the application, an important methodological discovery was made during the packaging process, involving conducting comparisons between two methods capable of turning web technology tools into standalone applications. These methods are HTML Application (HTA) and Progressive Web Application (PWA). Exploratory testing indicated inherent flaws with the conventional method involving HTA, including its inability to process modern JavaScript libraries and CSS frameworks efficiently. It was found that with its reliance on Internet Explorer's rendering engine, the performance of the ml5.js machine learning library and dynamic CSS attributes, necessary for real-time theme switching, is adversely affected in the HTA software framework.

On the other hand, the PWA strategy was much more resilient and provided native support for modern web standards with compatibility across multiple platforms. The above comparison evaluation confirmed the choice of PWA architecture as technically superior to other alternatives in complex client-side processing and adaptation applications.

Evaluation

Although the prototype achieves the main goal of changing the user interface based on the user's emotions, it was only possible to partially explore several enhancement ideas due to the project timeline. Currently, the solution mainly revolves around colour changes, however, the design could be more complex emotional design patterns for the interface. For example, the usage of background patterns reflecting different emotional states. At the same time, the dynamic typography changes where the font weight, spacing, or style is slightly changed according to the user's mood could be an additional reinforcement without a user being visually overloaded.

A significant functional limitation can be found in music management where the system is changing folders based on emotion detection during the upload process which is conflicting with the user manual organisation. Currently if want to organise music folder across different emotion categories and the system does not

allow to do so as it is automatically changing folders by emotion. A better version of the system would have a "music management mode" allowing the temporary stopping of folder changes so that it can freely add song to music folders without the emotion detection system interrupting them. This can be achieved by a simple toggle on the user interface or by detecting a longer interaction with the folder panel. These improvements could result in a better user experience.

Conclusions

This project makes it clear that real-time emotion detection is capable of being incorporated into responsive web interfaces. With face api from ml5.js, combined with the addition of CSS theming, it is obvious that a responsive system can adapt to emotions, all within a web browser. The PWA application allows for a method by which this adaptability can interact with devices, rather than a traditional application. To summarize, this project allows for responsive digital interfaces that can adapt to human emotions.

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